

1. Big picture – what this chapter is about

This week answers:

- How do we **prove** that “Design A is better than Design B” instead of just saying “I feel like it”?
- How do we design a **proper HCI experiment**?
- How do we avoid being fooled by random chance?

We’ll go through:

- Types of behavioral research
 - Hypotheses (null vs alternative)
 - IVs, DVs, typical ones in HCI
 - Components of an experiment & randomization
 - Significance tests, Type I & II errors, alpha, beta, power
 - Limitations of experiments
 - Then: experimental **design** (between/within, factorial, reliability, lifecycle, procedures)
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2. Types of behavioral research

Three main types:

2.1 Descriptive investigations

- Goal: **describe what is happening**.
- No attempt to prove cause-and-effect.
- Example:

- Watching users in a lab and writing down “how many times they click back”, “where they get stuck”.
- Doing an observation of how many people use their phone while walking.

You’re basically saying: “Here is an accurate picture of the behavior.”

👉 **MEMORIZE:** Descriptive = just describing, no cause-effect claim.

2.2 Relational investigations

- Goal: **find relationships** between variables.
- Example:
 - Does **age** relate to **error rate**?
 - Does **experience with computers** relate to **speed** on an app?

You might find: “Older users tend to have more errors.”

But:

- Relational studies **rarely show causation**.
- You can say “they are related”, but not “X causes Y”.

👉 **MEMORIZE:** Relational = finds correlation/relationship, usually **cannot state causation**.

2.3 Experimental research

- Goal: **establish causal relationships**.
- This is the strongest type if done properly.

Here we:

- Manipulate some variable (IV)
- Measure an outcome (DV)
- Use statistics to decide if changes in IV **caused** changes in DV.

Example:

“Does using a dark theme cause fewer visual fatigue complaints than light theme?”

👉 **MEMORIZE:** Experimental research = designed to establish **cause-and-effect**.

3. Research hypotheses

Experiments start with a **research hypothesis**.

- A **hypothesis** is a **precise problem statement** that can be tested in an empirical study.
- It's smaller and more focused than a big theory. One experiment usually tests one hypothesis.

Example in HCI:

“Using a touchscreen will result in **faster target selection** than using a mouse.”

That's something we can actually test.

👉 **MEMORIZE:** Hypothesis = precise, testable statement, usually for one experiment.

4. Null and alternative hypotheses

In statistics, we always set up **two competing statements**:

4.1 Null hypothesis (H_0)

- Usually: “**No difference**” between treatments.
- Example:
“There is **no difference** in task completion time between mouse and touchscreen.”

This is like the “default” or “status quo”.

4.2 Alternative hypothesis (H_1 or H_a)

- Opposite of the null.
- There **is** a difference (or a specific direction of difference).
- Example:

“Touchscreen is **faster than** mouse for target selection.”

The goal of our experiment:

Find **statistical evidence** to **reject the null hypothesis**, so we can support the alternative.

Also, a good hypothesis must **specify**:

- The **independent variables** (IVs)
- The **dependent variables** (DVs)

👉 MEMORIZE:

- H_0 = no difference.
- H_1 = difference (or specific effect).
- Experiment tries to **reject H_0** to support H_1 .

5. Independent and dependent variables

5.1 Independent Variable (IV)

- The factor that we **manipulate**.
- It is “independent” of the participant’s behavior.

- Often called the **cause** (although we only say that after results).
- Examples in HCI:
 - Type of device (mouse vs touchscreen vs gesture)
 - Type of design (flat UI vs skeuomorphic)
 - User group (novice vs expert)
 - Environment (quiet vs noisy)

5.2 Dependent Variable (DV)

- The **outcome** we measure.
- It **depends** on the participant's behavior and the IV.
- Examples in HCI:
 - Task completion time (seconds)
 - Error rate
 - Satisfaction rating (Likert scale)
 - Number of mistakes
 - NASA-TLX workload score

👉 MEMORIZE:

- IV: what we **control/manipulate**
- DV: what we **measure**

6. Typical IVs and DVs in HCI

6.1 Typical IVs in HCI

1. **Technology-related:**

- Type of device (mouse, touch, stylus, gesture)
- Type of interface (menu vs toolbar vs ribbon)
- Type of feedback (audio vs visual vs haptic)

2. **User-related:**

- Age, gender
- Computer experience
- Professional domain (e.g., doctors vs students)
- Education level
- Culture
- Motivation / mood
- Disabilities (visual, physical, cognitive)

3. **Context-of-use related:**

- Physical environment (noise, lighting, sitting vs walking)
- User status (tired vs fresh, stressed vs relaxed)
- Social status / whether others are present (alone vs with peers/supervisors)

6.2 Typical DVs in HCI

Common dependent variables:

- **Efficiency** → e.g., **task completion time**, speed
- **Accuracy** → e.g., **error rate**, number of mistakes
- **Subjective satisfaction** → e.g., **Likert-scale ratings** ("1–7 how satisfied are you?")

- **Ease of learning / retention** → how quickly users learn, how much they remember later
- **Physical or cognitive demand** → e.g., **NASA TLX** workload

👉 **MEMORIZE at least 3 typical DVs** (time, error rate, satisfaction) and **2–3 IV categories** (technology/user/context).

7. Components of an experiment

An experiment has 3 basic components:

1. **Treatments / conditions**

- Different techniques/devices/procedures we compare.
- Example: Condition 1 = mouse, Condition 2 = touchscreen, Condition 3 = gesture.

2. **Units**

- The objects/subjects to whom we apply treatments.
- In HCI, units are typically **human participants** with specific characteristics (e.g., age, gender, experience).

3. **Assignment method**

- How we assign participants to conditions.
- Example: randomization (randomly assigning people to mouse vs touch conditions).

👉 **MEMORIZE: Treatments, Units, Assignment method.**

8. Randomization

Randomization = randomly assigning treatments to participants.

- In a totally randomized experiment, **no one (even the researcher)** can predict who gets which condition.
- Purpose:
 - Reduce **bias** (e.g., not putting all experts in one group).
 - Make groups comparable on average.

Methods:

- Using a **random number table**
 - Using **software** (Excel, SPSS, R, online tools)
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9. Significance tests – why we need them

If we could test **every single person in the population**, we'd know the true difference exactly
→ no need for statistics.

But in real life:

- Population is large (all smartphone users, all students, etc.).
- We only test a **sample**.

Significance tests help us answer:

“Is the difference we see in our **sample** likely to be real in the **whole population**, or just random chance?”

So:

- We compute a **p-value**.
- We compare it to a **significance level (alpha)** (like 0.05).
- If $p < \alpha$ → we say the result is **statistically significant** and reject H_0 .

👉 **MEMORIZE:** We use significance tests because we only test a **sample**, not the whole population.

10. Type I and Type II errors

When we do hypothesis testing, we can go wrong in **two ways**:

10.1 Type I error (α error / false positive)

- We **reject the null** when it is actually **true**.
- In plain English:
 - We say “there **is** a difference” when **in reality there is NO difference**.
- Called “gullibility” – we believed an effect that wasn’t there.

Example:

- We conclude “touchscreen is faster than mouse”, but actually they are the same and your result was just random.

10.2 Type II error (β error / false negative)

- We **do not reject the null** when it is actually **false**.
- In plain English:
 - We say “no difference” but actually **there is a real difference**.
- Called “blindness” – we fail to see an effect that exists.

Example:

- There **is** a better UI, but your study missed it due to small sample or noisy data.

👉 **MEMORIZE:**

- Type I = false positive / reject H_0 wrongly / gullibility.
- Type II = false negative / fail to reject H_0 wrongly / blindness.

It is generally believed that **Type I errors are worse**, because:

- They may lead us to adopt a **bad treatment** that is actually not better (or worse than current).

11. Alpha, beta, and power

- **Alpha (α):**
 - Probability of **Type I error**.
 - Commonly set to **0.05** in HCI and many fields.
- **Beta (β):**
 - Probability of **Type II error**.
- **Power = $1 - \beta$**
 - Probability of **correctly rejecting** a false null hypothesis.
 - In other words, chance of **detecting a real effect**.

Relationship:

- If you **decrease alpha** (make it stricter, e.g., from 0.05 to 0.01), you reduce Type I errors but **increase** Type II errors (harder to detect real effects).
- There is a **trade-off**.

Because Type I is considered worse, the standard alpha = **0.05** is widely used to control Type I errors.

👉 **MEMORIZE:**

- $\alpha = P(\text{Type I}), \beta = P(\text{Type II}), \text{Power} = 1 - \beta$.
 - Lower $\alpha \rightarrow$ fewer Type I, more Type II.
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12. Limitations of Experimental Research

Experimental research is powerful, but:

1. Needs **well-defined, testable hypotheses**
 - You must limit to a **small number of IVs and DVs**.
 - Real life is messy, but experiments must be focused.
2. Needs **strict control** of other factors that might influence the DV
 - Lighting, noise, prior practice, etc.
 - Hard to control everything perfectly.
3. **Lab experiments may not represent real-world behavior**
 - People behave differently when being watched in a lab vs using a system naturally at home.

👉 **MEMORIZE: 3 limitations:** needs clear hypotheses, strict control, lab $\neq$ real world.

13. Experimental Design – Chapter 3 stuff

Now we move from “what is an experiment” $\rightarrow$ “how to design one”.

We talk about:

- True experiments vs quasi vs non-experiments
- Between-group vs within-group designs

- Factorial designs
 - Reliability & errors
 - Lifecycle & session procedures
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14. Three groups of studies

- **Experiments** – full control, randomization, hypotheses, significance tests.
- **Quasi-experiments** – some features of experiments but **less control** (e.g. no full randomization).
- **Non-experiments** – observational or correlational (like surveys, field observations).

👉 From the slides, the focus is on **true experiments**: the “gold standard”.

15. Characteristics of a true experiment

A true experiment typically:

1. Is based on at least **one hypothesis**
2. Has **multiple conditions** (e.g., interface A vs B vs C)
3. Has a DV that can be **quantitatively measured**
4. Uses **statistical significance tests**
5. Tries to **remove biases** (through randomization, control)
6. Is **replicable** (someone else can repeat it and get similar results)

👉 **MEMORIZE this list**; sir can easily ask: “What are characteristics of a true experiment?”

16. Factors to consider when designing an experiment

1. Research hypothesis

- Must be clear and testable.
- Not too vague; must specify IVs and DVs.

2. Dependent variables

- Must be **easy to measure** and appropriate.

3. Independent variables / conditions

- Must be **easy to control**.
 - Example: you can control UI design, but you can't easily control a user's mood.
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17. Example hypothesis from slides

"There is no difference between the target selection speed when using a mouse, a touchscreen, or a gesture to select icons of different sizes (small, medium, or large)."

Here:

- IVs:
 - Type of input: 3 levels (mouse, touchscreen, gesture)
 - Icon size: 3 levels (small, medium, large)
- Total conditions = $3 \times 3 = 9$ conditions.

DV:

- Target selection speed (time).

This is an example of **two IVs**, each with 3 levels → factorial design later.

18. Basic design structure

Two key questions:

1. How many **independent variables**? (1, 2, 3...)
2. How many **levels** does each IV have? (e.g., 2 device types, 3 icon sizes)

This determines:

- Number of conditions
 - Complexity of experiment
 - Sample size needed
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19. Investigating one IV – Between-group vs Within-group

19.1 Between-group design (between-subject)

- Each participant experiences **only ONE condition**.
 - Example: Participant A → mouse only
 - Participant B → touchscreen only
 - No one uses more than one type.

Advantages:

- Cleaner, better control of **learning effect**
 - They don't get practice with multiple conditions.
- **Shorter time** for each participant

- Less **fatigue** and frustration

Disadvantages:

- **Individual differences** affect results (some people are naturally fast/slow).
 - Harder to detect differences between conditions (more noise).
 - Requires **larger sample size** (need many participants per condition).
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19.2 Within-group design (within-subject)

- Each participant experiences **multiple (or all) conditions**.
 - Example: the same person does tasks with mouse, then touchscreen, then gesture.

Advantages:

- Requires **smaller sample size** (each person gives data for all conditions).
- Easier to detect differences between conditions because **each person is their own control**.

Disadvantages:

- **Learning effect:**
 - Participants may get better just from practice, not because the interface is better.
 - Takes **longer time** for each participant (they do more tasks).
 - More **fatigue and frustration**.
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19.3 When to use which?

Between-group is better when:

- Tasks are **simple**
- Learning effect would **strongly impact** performance
- Within-group design is impossible (e.g., once someone sees condition A, they are biased for condition B)

Within-group is better when:

- Learning effect is **small or can be controlled** (e.g., counterbalancing order)
- You have a **small participant pool**
- You want high sensitivity with fewer participants

👉 **MEMORIZE:** definition + key pros/cons + when to use between vs within.

20. More than one IV – Factorial design

When you have **multiple IVs**, you often use **factorial design**.

- You divide experiment conditions into subsets based on combinations of IV values.
- You don't just study each IV separately; you can also study **interaction effects**.

Number of conditions:

- Simply: multiply levels.
 - Example: 3 input methods × 3 icon sizes = 9 conditions.

Types of factorial design:

- Between-group design (each condition uses different participants)
- Within-group design (each participant goes through all combinations)
- Split-plot design (some IVs between, some within)

Interaction effect

“The differing effect of one IV on the DV, depending on the level of another IV.”

Example:

- Maybe gesture is only slower than mouse when icons are small, but similar speed when icons are large.
- This means “effect of device type” depends on “icon size”.

👉 **MEMORIZE: Interaction effect definition.**

21. Reliability of experiment results

Two main kinds of errors affecting reliability:

21.1 Random errors (chance errors / noise)

- Variations in both directions (sometimes up, sometimes down).
- Occur **by chance**.
- Example: someone had a headache one day, random misclicks.
- Can be **reduced** by:
 - Using larger **sample size**, averaging across people.

21.2 Systematic errors (biases)

- Always push the actual value in the **same direction**.
- Example:
 - A clock that is always 5 minutes slow.
 - An experimenter always giving more hints to one group.

- Cannot be removed just by increasing sample size; they remain.

👉 **MEMORIZE: Random error vs systematic error; systematic = bias in one direction.**

21.3 Sources of systematic errors

Five major sources:

1. **Measurement instruments**

- Faulty stopwatch, inaccurate productivity tracker, miscalibrated scales.

2. **Experimental procedures**

- Different instructions to different groups; not standardizing training.

3. **Participants**

- Selection bias (e.g., all experts in one condition).

4. **Experimenter behavior**

- Tone, hints, body language favoring one condition.

5. **Experimental environment**

- One group tested in quiet lab, other in noisy hallway.

👉 **MEMORIZE these 5 sources.**

22. Lifecycle of an experiment

Typical steps:

1. **Identify a research hypothesis**

2. **Specify the design** of the study (IVs, DVs, between/within...)

3. Run a **pilot study** (small test) to check design, system, instruments
 4. **Recruit participants**
 5. Run **actual data collection sessions**
 6. **Analyze data** (statistics)
 7. **Report results**
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23. Experiment session procedure

For each participant:

1. Ensure **systems/devices/instruments** are ready
2. **Greet** the participant
3. **Introduce purpose** and procedures (without biasing them)
4. Get **informed consent**
5. Assign participant to a condition using **pre-defined randomization**
6. Participant completes **training tasks**
7. Participant completes **actual tasks**
8. Participant answers **questionnaires** (e.g., satisfaction, NASA TLX)
9. **Debriefing** session (explain any deception / purpose more fully)
10. **Payment** or thanks (if any)

👉 **MEMORIZE** this sequence in rough order.

2. Practice Section

Now: 5 MCQs, 5 Short Questions, 1 Case Study – all with answers & explanations.

A. MCQs

MCQ 1

Which of the following **best** describes a **research hypothesis**?

- A. A broad theory that explains many phenomena
- B. A precise, testable statement that can be examined by a single experiment
- C. A description of observations without any prediction
- D. A list of variables that cannot be measured

Answer: B

Explanation:

A hypothesis is **narrow and testable**, usually addressing a specific problem that a single experiment can investigate, unlike a broad theory.

MCQ 2

In an HCI experiment comparing mouse vs touchscreen, the **input device type** is:

- A. A dependent variable
- B. A random variable
- C. An independent variable
- D. A systematic error

Answer: C

Explanation:

The researcher **manipulates** the device type, so it is the **independent variable**. Task completion time or error rate would be dependent variables.

MCQ 3

Which statement about **Type I and Type II errors** is correct?

- A. Type I error = failing to reject a false null hypothesis
- B. Type II error = rejecting a true null hypothesis
- C. Type I error is often considered more serious than Type II error
- D. Reducing alpha decreases both Type I and Type II errors simultaneously

Answer: C

Explanation:

Type I = rejecting a **true** null (false positive, "gullibility").

Type II = failing to reject a **false** null (false negative, "blindness").

Type I is often considered worse; decreasing alpha reduces Type I but increases Type II.

MCQ 4

In a **between-group design**:

- A. Each participant experiences all conditions
- B. Each participant experiences only one condition
- C. Randomization is not required
- D. Learning effects are always higher

Answer: B

Explanation:

Between-group (between-subject) design assigns each participant to **only one condition**. Learning effects are lower; randomization is still important.

MCQ 5

Which of the following is an example of a **systematic error** source?

- A. Participant misclicks due to random distraction
- B. Measuring task time with a stopwatch that always starts 1 second late
- C. Some participants are naturally faster typers
- D. Having slightly different internet speeds each minute

Answer: B

Explanation:

The stopwatch that is always late introduces a **consistent bias** in the same direction → systematic error. Random distractions and natural differences are part of random error.

B. Short Answer Questions

SAQ 1

Differentiate between descriptive, relational, and experimental research in one or two lines each.

Model answer:

- **Descriptive:** aims to **describe** behavior or phenomena accurately, without claiming cause.
 - **Relational:** examines **relationships** or correlations between variables but usually **cannot show causation**.
 - **Experimental:** manipulates one or more **independent variables** to see their causal effect on **dependent variables**, using controlled conditions and statistics.
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SAQ 2

What is a null hypothesis? Give an HCI example.

Model answer:

A null hypothesis usually states that there is **no difference** between experimental conditions. Example: "There is no difference in average task completion time between the old UI and the new UI." We try to collect evidence to **reject** this statement in favor of an alternative.

SAQ 3

Explain the main advantages and disadvantages of a within-group design.

Model answer:

In a within-group design, each participant experiences **multiple conditions**. The main advantages are that it requires **fewer participants** and makes it easier to detect differences because each participant acts as their own control. The main disadvantages are strong **learning effects**, longer sessions, and more **fatigue or frustration** for participants.

SAQ 4

What is an interaction effect in a factorial experiment? Provide a simple example.

Model answer:

An interaction effect occurs when the effect of one independent variable on the dependent variable **depends on the level of another IV**. For example, in an HCI experiment, gesture input might be slower than mouse when icons are **small**, but similar speed when icons are **large**. So the effect of “input type” changes depending on “icon size”.

SAQ 5

List any three steps from the lifecycle of an experiment and briefly explain them.

Model answer:

Three steps are:

1. **Identify a research hypothesis** – decide the specific question and prediction.
 2. **Run a pilot study** – try a small test run to check if tasks, instructions, and instruments work.
 3. **Analyze data** – apply statistical tests to decide whether to reject the null hypothesis and interpret what the results mean.
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C. Case Study Question

Case Study

You want to test whether a **gesture-based interface** on a smart TV allows users to **select movies faster** than a standard **remote control** interface.

You design a lab study where:

- Participants are randomly assigned to either **gesture** or **remote** condition (each participant uses only one).
- Everyone has to select 20 specific movie titles.

- You record the **total time** taken and number of **selection errors**.

Question:

Using Week 8 concepts, answer:

1. What kind of research is this (descriptive, relational, experimental)?
 2. Clearly state a suitable **null** and **alternative** hypothesis.
 3. Identify the **IV**, **DVs**, and the **type of design** (between or within).
 4. Mention one possible **Type I error** and one **Type II error** in this study.
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Model Answer (structured)

1. **Type of research**

This is **experimental research**, because we **manipulate** the interface type (gesture vs remote) and measure outcomes (time, errors) to test causal effects.

2. **Hypotheses**

- **Null hypothesis (H_0):**
There is **no difference** in movie selection time between the gesture-based interface and the remote-control interface.
- **Alternative hypothesis (H_1):**
The gesture-based interface allows **faster** movie selection than the remote-control interface.
(You could also phrase H_1 as “there is a difference in time between the two interfaces” if you want a non-directional alternative.)

3. **Variables and design type**

- **Independent Variable (IV):**
Type of interface: Gesture vs Remote.
- **Dependent Variables (DVs):**
 - Total time taken to select 20 movies (efficiency).

- Number of selection errors (accuracy).
- **Design type:**
This is a **between-group** (between-subject) design because **each participant experiences only one condition** (either gesture or remote).

4. Type I and Type II errors in this context

- **Type I error:**
You conclude that **gesture is faster** than remote (reject H_0), when in reality there is **no true difference**. You might deploy a more complex gesture system for no real benefit.
- **Type II error:**
You conclude there is **no difference** (fail to reject H_0), when in reality gesture is **actually faster**. You miss an opportunity to adopt a better interface.

You could extend this answer by discussing randomization (to reduce bias), potential systematic errors (e.g., different room setup for each group), and the need for an appropriate sample size to achieve good **statistical power**.